

Waseem Raza, PhD

Machine Learning Engineer | Generative & Agentic AI

LLM Applications | RAG | Agent Orchestration | Cloud Data Pipelines
New Jersey, United States ☎ +1 732-803-9703 ✉ waseem.placements@gmail.com | waseemraza844@gmail.com
LinkedIn | GitHub | Google Scholar | Portfolio

Professional Summary

Generative AI and Machine Learning Engineer with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience. Build LLM applications, retrieval-augmented generation (RAG) pipelines, structured information-extraction workflows, tool-using agents, and AI-ready data pipelines using Python, LangChain, LangGraph, Hugging Face, OpenAI APIs, Azure Databricks, and cloud services. Experience includes multimodal diagnostic prototypes, prompt engineering, embeddings, semantic retrieval, agent orchestration, model evaluation, domain adaptation, and production-oriented AI delivery.

Technical Skills

Generative & Agentic AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering, document ingestion, chunking, embeddings, semantic retrieval, structured information extraction, tool-using agents, agent orchestration, and grounded response generation

Microsoft AI Ecosystem: Microsoft Foundry, Azure OpenAI, Azure AI Services, Azure AI Search, Azure Machine Learning, Foundry models and agents, Azure Content Understanding, Semantic Kernel, and responsible AI concepts

LLM Frameworks & Models: Hugging Face Transformers, LangChain, LangGraph, OpenAI APIs, transformer architectures, fine-tuning concepts, NLP foundation models, prompt templates, structured outputs, and LLM evaluation

Data & AI Platforms: Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Blob Storage, Azure Data Lake, Google BigQuery, and cloud analytics workflows

Data Engineering: ETL/ELT pipelines, AI-ready datasets, batch processing, ingestion, cleansing, schema transformation, validation, feature engineering, data-quality assessment, and structured and unstructured data workflows

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, transfer learning, domain adaptation, adversarial learning, anomaly detection, classification, regression, time-series forecasting, cross-validation, and model evaluation

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, shell scripting, model deployment, reproducible workflows, and cloud-native application development

Programming & Analytics: Python, SQL, C/C++, and R; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter, and exploratory data analysis

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

02/2025 - Present

Bedminster, NJ

- Lead end-to-end development and validation of AI-driven E911 diagnostic workflows, combining NLP, domain-adaptive machine learning, and production-oriented data pipelines for critical service assurance.
- Develop multimodal GenAI-assisted diagnostic prototypes that combine work-order text, location information, network context, and operational signals for VoNR and 5G service-assurance use cases.
- Design reusable model-validation, error-analysis, and failure-mode assessment workflows to compare performance across heterogeneous operational domains and distribution shifts.
- Partner with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and deployable data assets.

Data Scientist and AI/ML Engineer

Impac Exploration Services

09/2024 - 12/2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data using Python and Azure-based data services.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Collaborated with geologists to translate domain requirements into AI-ready datasets and structured analytical outputs.

Advanced-Degrees Intern <i>AT&T Labs</i> - Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%. - Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation. - Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.	06/2024 - 08/2024 <i>Austin, TX</i>
Advanced Wireless Technology Development Intern <i>Futurewei Technologies</i> - Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets. - Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions. - Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.	09/2022 - 01/2023 <i>Schaumburg, IL</i>
Advanced-Degrees Intern <i>AT&T Labs</i> - Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%. - Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership. - Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.	06/2022 - 08/2022 <i>Bedminster, NJ</i>
Graduate Research Assistant <i>AI4Networks, University of Oklahoma</i> - Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed. - Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling. - Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation. - Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%. - Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.	09/2019 - 07/2025 <i>Tulsa, OK</i>
Education	
PhD, Electrical and Computer Engineering University of Oklahoma, Norman, Oklahoma, USA	09/2019 - 07/2025
MSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	08/2014 - 08/2016
BSc, Telecommunication Engineering University of Engineering and Technology, Taxila, Pakistan	11/2010 - 06/2014
Selected Projects	
Agentic AI and RAG Applications <i>Portfolio Project</i> Practical experiments with retrieval, tool-using agents, orchestration, memory, evaluation, and multi-agent workflows. <i>Tools & methods:</i> LangGraph, CrewAI, AutoGen, RAG, Vector Databases	2026 / In progress
Automated Meeting Minutes Generator with LLMs <i>Guided Project</i> A workflow that transcribes meetings, produces structured summaries, and extracts decisions and action items for downstream collaboration. <i>Tools & methods:</i> Python, Speech-to-Text, LLM Summarization, Workflow Automation	2025
LLM-Based Extraction from Unstructured Documents <i>Guided Project</i> Document intelligence pipeline for extracting entities, fields, and tables from reports, invoices, and contracts. <i>Tools & methods:</i> LLMs, OCR, Named Entity Recognition, Data Integration	2025
AI-Ready Lakehouse Pipeline <i>Portfolio Project</i> An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads. <i>Tools & methods:</i> Azure, Databricks, Spark, Data Engineering	2024
System Design Case-Study Portfolio <i>Guided Project</i> Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs. <i>Tools & methods:</i> System Design, APIs, Distributed Systems, Scalability	2026 / In progress

Certifications

1. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

Focus: Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

Tools & Skills: Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development

2. IBM Generative AI Engineering Professional Certificate — IBM / Coursera | Completed (13/16) courses

Focus: Sixteen-course professional certificate developing job-ready skills across AI foundations, Python application development, machine learning, deep learning, NLP, transformer architectures, LLM fine-tuning, RAG, LangChain, and AI agents.

Tools & Skills: Generative AI Engineering, Large Language Models, Transformer Architectures, Prompt Engineering, RAG, LangChain, AI Agents, Hugging Face, PyTorch, Keras, Scikit-learn, Python, Flask, Gradio, PEFT, LoRA, QLoRA, RLHF, DPO, PPO, NLP, Machine Learning, Deep Learning, Vector Databases

3. IBM RAG and Agentic AI Professional Certificate — [IBM / Coursera](#) | Completed (8/8) courses | 02/2026 | [Certificate](#)

Original eight-course credential.

Focus: Completed eight-course professional certificate covering retrieval-augmented generation, vector databases, multimodal applications, and agentic AI workflows.

Tools & Skills: Retrieval-Augmented Generation, Agentic AI, LangChain, LangGraph, Vector Databases, ChromaDB, FAISS, LlamaIndex, Multimodal AI, Tool Calling, Prompt Engineering, CrewAI, AutoGen / AG2, BeeAI, Gradio, Python, Flask

4. Generative AI Fundamentals Specialization — [IBM / Coursera](#) | Completed (5/5) courses | 07/2026 | [Certificate](#)

Focus: Five-course foundation in generative AI, prompt engineering, LLMs, and practical enterprise applications.

Tools & Skills: Generative AI, Prompt Engineering, Foundation Models, Responsible AI, IBM watsonx, Hugging Face, Business Applications